

처음 만나는 디지털 논리회로

제06장. 논리식의 간소화

학습목표

- 카르노 맵을 이용해 논리식을 간소화할 수 있다.
- 모든 게이트들을 NAND와 NOR 게이트로 나타내는 방법을 이해하고 이를 응용할 수 있다.
- XOR 게이트와 XNOR 게이트의 특징을 이해하고 이를 활용할 수 있다.

6.1 2변수 카르노 맵

6.2 3변수 카르노 맵

6.3 4변수 카르노 맵

6.4 선택적 카르노 맵

6.5 논리식의 카르노 맵 작성

6.6 5변수, 6변수 카르노 맵

6.7 NAND와 NOR 게이트로의 변환

6.8 XOR와 XNOR 게이트

□ 개요

- 불 대수를 이용한 간소화하는 방법은 복잡하고 검증도 어렵다.
- 카르노 맵(1953년 Maurice Karnaugh가 소개)을 이용하면 논리식을 쉽게 간소화할 수 있다.

□ 2변수 카르노 맵 표현 방법

$\begin{array}{c} \diagdown \\ B \\ A \end{array}$	$\bar{B}$	B
$\bar{A}$	$\bar{A}\bar{B}$	$\bar{A}B$
A	$A\bar{B}$	AB

$\begin{array}{c} \diagdown \\ B \\ A \end{array}$	$\bar{B}$	B
$\bar{A}$	m_0	m_1
A	m_2	m_3

$\begin{array}{c} \diagdown \\ B \\ A \end{array}$	0	1
0	0	1
1	2	3

$\begin{array}{c} \diagdown \\ A \\ B \end{array}$	$\bar{A}$	A
$\bar{B}$	m_0	m_2
B	m_1	m_3

- 무관항(don't care): 입력이 결과에 영향을 미치지 않는 최소항
- x 로 표시하거나 d로 표시한다.

□ 일반항과 무관항 표현

$A \backslash B$	0	1
0	1	
1		1

m_0 (points to cell (0,0))
 m_3 (points to cell (1,1))

$$F(A, B) = \sum m(0, 3)$$

$A \backslash B$	0	1
0	1	x
1		1

d_1 (points to cell (0,1))

$$F(A, B) = \sum m(0, 3) + \sum d(1)$$

- 출력이 1이거나 무관항만 표시한다.
- 출력 0을 표시하여도 되지만 일반적으로 생략한다.

□ 카르노 맵을 이용한 간소화 방법

- ① 출력이 같은 항을 1, 2, 4, 8, 16개로 그룹을 지어 묶는다.
- ② 바로 이웃한 항들끼리 묶는다.
- ③ 반드시 직사각형이나 정사각형의 형태로 묶어야 한다.
- ④ 최대한 크게 묶는다.
- ⑤ 중복하여 묶어서 간소화된다면 중복하여 묶는다.
- ⑥ 무관항의 경우 간소화될 수 있으면 묶어 주고, 그렇지 않으면 묶지 않는다.

$A \backslash B$	0	1
0	1	1
1		

불 대수의 법칙으로 풀면

$$\begin{aligned}
 F &= \overline{A}\overline{B} + \overline{A}B \\
 &= \overline{A}(\overline{B} + B) = \overline{A} \cdot 1 = \overline{A}
 \end{aligned}$$

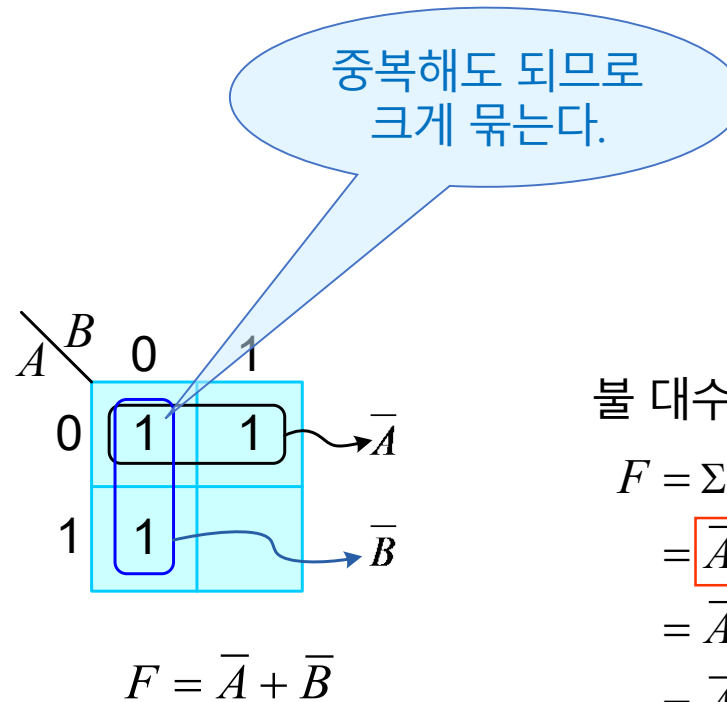
$A=0$ 이므로 $\overline{A}$

$B=0$ and 1 이므로 제거

즉, 한 변수에서 서로 다른 값이 묶여지면 제거한다.

□ 간소화 예

입력		출력
A	B	F
0	0	1
0	1	1
1	0	1
1	1	0



불 대수의 법칙으로 풀면

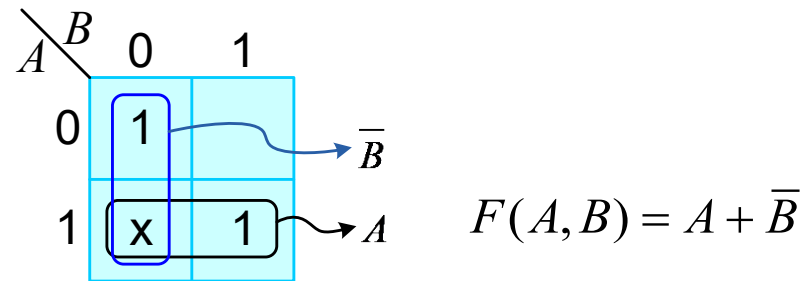
$$\begin{aligned}
 F &= \Sigma m(0,1,2) = \bar{A}\bar{B} + \bar{A}B + A\bar{B} \\
 &= \boxed{\bar{A}\bar{B}} + \bar{A}B + \boxed{A\bar{B}} + A\bar{B} \\
 &= \bar{A}(\bar{B} + B) + \bar{B}(\bar{A} + A) \\
 &= \bar{A} \cdot 1 + \bar{B} \cdot 1 \\
 &= \bar{A} + \bar{B}
 \end{aligned}$$

예제 6-1

다음 논리식을 카르노 맵을 이용해 간소화하라. 단, d 는 무관항이다.

$$F(A, B) = \sum m(0, 3) + \sum d(2)$$

풀이



End of Example

6.2 3변수 카르노 맵

3변수 카르노 맵

□ 3변수 카르노 맵 표현 방법

$\backslash BC$	$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
$\overline{A}$	$\overline{A}\overline{B}\overline{C}$	$\overline{A}\overline{B}C$	$\overline{A}B\overline{C}$	$\overline{A}BC$
A	$A\overline{B}\overline{C}$	$A\overline{B}C$	$AB\overline{C}$	ABC

$\backslash BC$	00	01	11	10
0	0	1	3	2
1	4	5	7	6

$\backslash C$	$\overline{C}$	C
$\overline{A}B$	$\overline{A}\overline{B}\overline{C}$	$\overline{A}B\overline{C}$
$\overline{A}\overline{B}$	$\overline{A}\overline{B}C$	$\overline{A}B\overline{C}$
AB	$AB\overline{C}$	ABC
$A\overline{B}$	$A\overline{B}\overline{C}$	$A\overline{B}C$

$\backslash C$	0	1
00	0	1
01	2	3
11	6	7
10	4	5

$\backslash AB$	$\overline{A}\overline{B}$	$\overline{A}B$	AB	$A\overline{B}$
$\overline{C}$	$\overline{A}\overline{B}\overline{C}$	$\overline{A}B\overline{C}$	$AB\overline{C}$	$A\overline{B}\overline{C}$
C	$\overline{A}\overline{B}C$	$\overline{A}BC$	ABC	$A\overline{B}C$

$\backslash AB$	00	01	11	10
0	0	2	6	4
1	1	3	7	5

행과 열을 바꾸어도 상관없다.
설계자가 선호하는 방법을 선택하면 된다.

6.2 3변수 카르노 맵

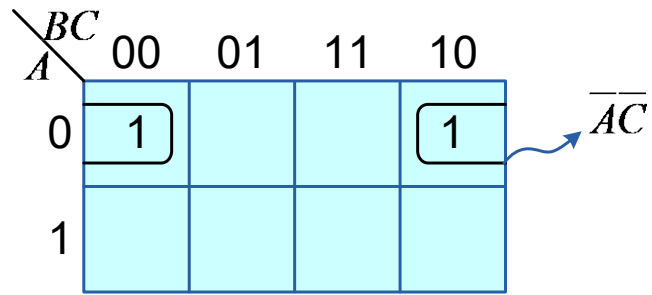
3변수 카르노 맵

□ 간소화 예1 : $F(A, B, C) = \Sigma m(0, 1, 6, 7)$

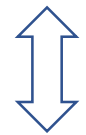
$\begin{array}{c} \diagup BC \\ \diagdown A \end{array}$	00	01	11	10	
0	1	1			$\overline{A}\overline{B}$
1			1	1	AB

$$F = \overline{A}\overline{B} + AB$$

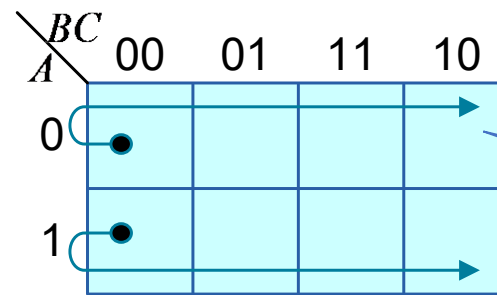
□ 간소화 예2 : 양쪽 끝 묶음



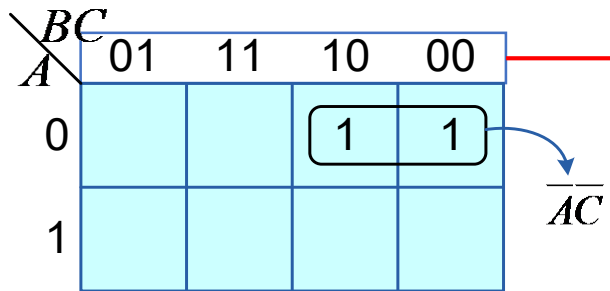
$$F = \overline{A}\overline{C}$$



동일한 카르노 맵



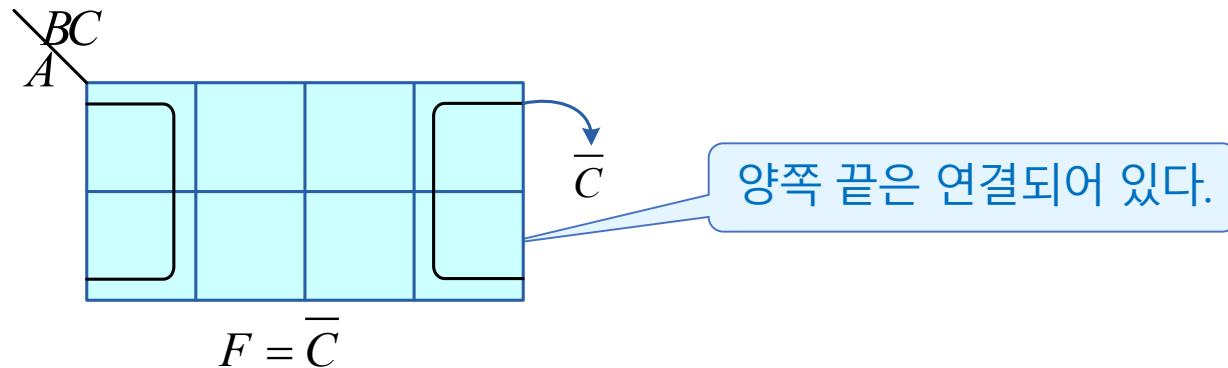
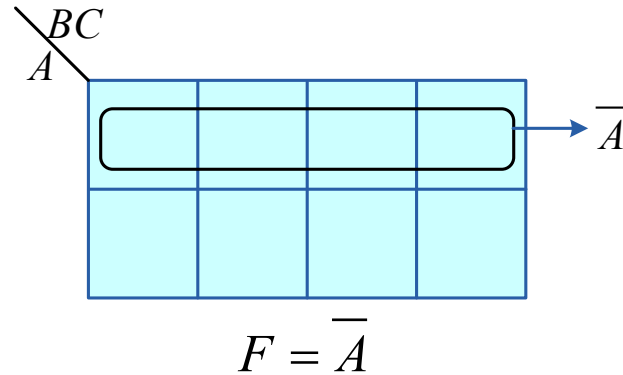
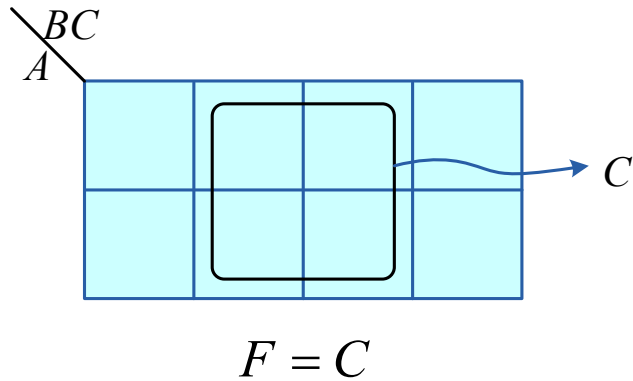
양쪽 끝은 연결
되어 있다.



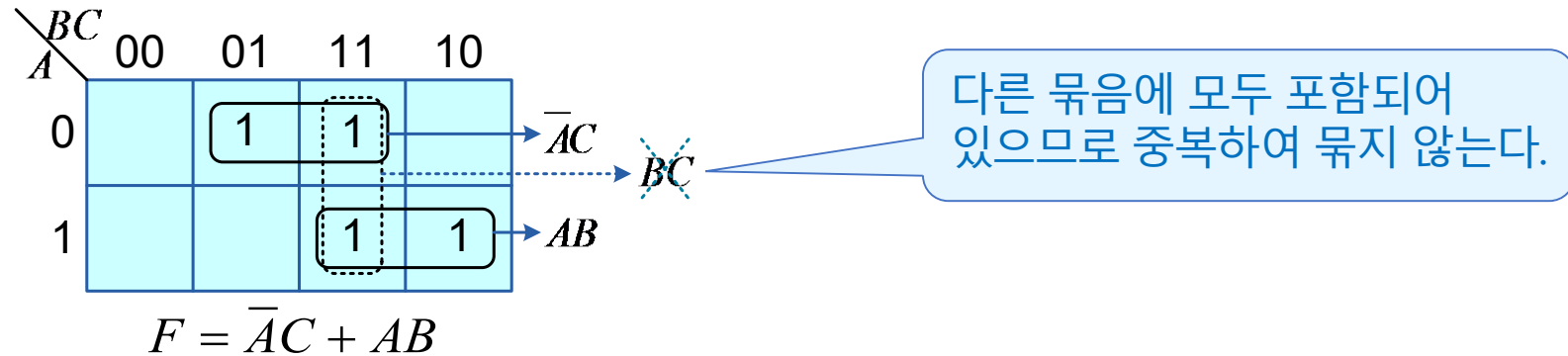
$$F = \overline{A}\overline{C}$$

이웃하는 비트들이 한 비트만 다르다면
순서는 관계없다.

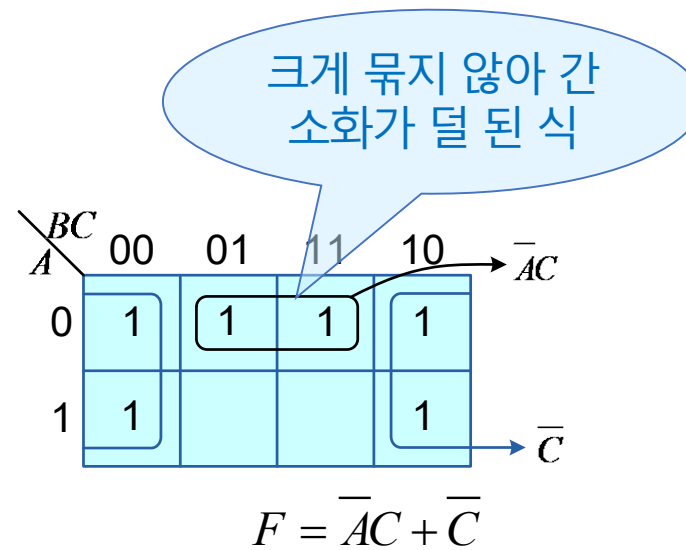
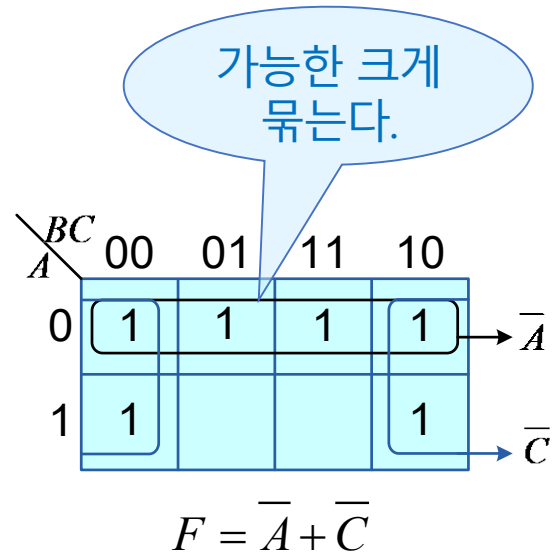
□ 간소화 예3 : 4개 항 묶음



□ 간소화 예4 : 다른 묶음에 모두 포함되어 있는 경우는 묶지 않는다.



□ 간소화 예5 : 최대한 크게 묶는다.



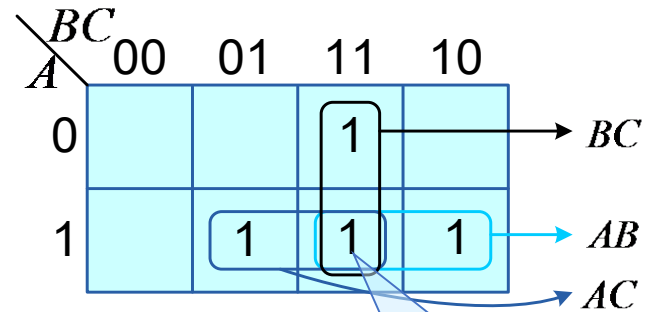
6.2 3변수 카르노 맵

3변수 카르노 맵

□ 간소화 예6 : 세 번 중복하여 묶는 경우

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$F = \sum m(3, 5, 6, 7) = AB + BC + AC$$



세 번 중복하여 묶은 경우

6.2 3변수 카르노 맵

3변수 카르노 맵

□ 간소화 예7 : 모두 0이거나 모두 1인 경우

$\backslash BC$	00	01	11	10
A 0				
A 1				

$F = 0$

모두 0이면 논리식은
 $F = 0$ 이다.

$\backslash BC$	00	01	11	10
A 0	1	1	1	1
A 1	1	1	1	1

$F = 1$

모두 1이면 논리식은
 $F = 1$ 이다.

예제 6-2

다음 3변수 카르노 맵에서 간소화된 논리식 F 를 구하라.

$AB \backslash C$	0	1
00	0	0
01	0	0
11	1	1
10	1	0

풀이

$AB \backslash C$	0	1
00	0	0
01	0	0
11	1	1
10	1	0

$\rightarrow AB$

$\rightarrow A\bar{C}$

$$F = AB + A\bar{C}$$

6.3 4변수 카르노 맵

4변수 카르노 맵

□ 4변수 카르노 맵 표현 방법

$AB \backslash CD$	00	01	11	10
00	$\overline{A}\overline{B}\overline{C}\overline{D}$	$\overline{A}\overline{B}\overline{C}D$	$\overline{A}\overline{B}C\overline{D}$	$\overline{A}\overline{B}CD$
01	$\overline{A}\overline{B}C\overline{D}$	$\overline{A}\overline{B}CD$	$\overline{A}B\overline{C}\overline{D}$	$\overline{A}B\overline{C}D$
11	$\overline{A}B\overline{C}\overline{D}$	$\overline{A}B\overline{C}D$	$AB\overline{C}\overline{D}$	$AB\overline{C}D$
10	$\overline{A}BC\overline{D}$	$\overline{A}BCD$	$A\overline{B}\overline{C}\overline{D}$	$A\overline{B}\overline{C}D$

$AB \backslash CD$	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$AB \backslash CD$	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

상하 좌우는
연결되어 있다.

TIP

$AB \backslash CD$	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$F = 1$

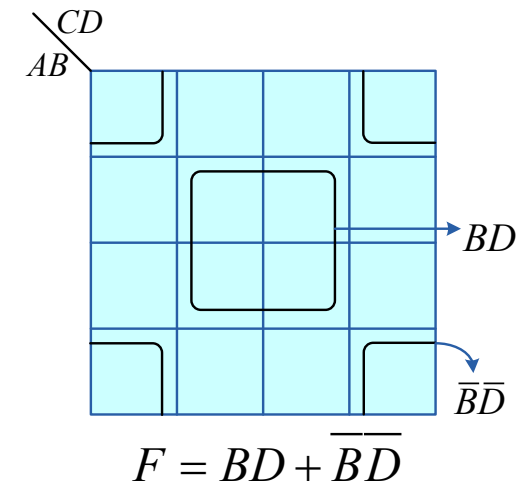
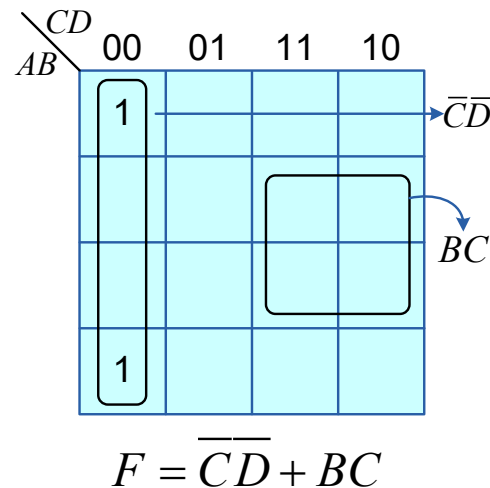
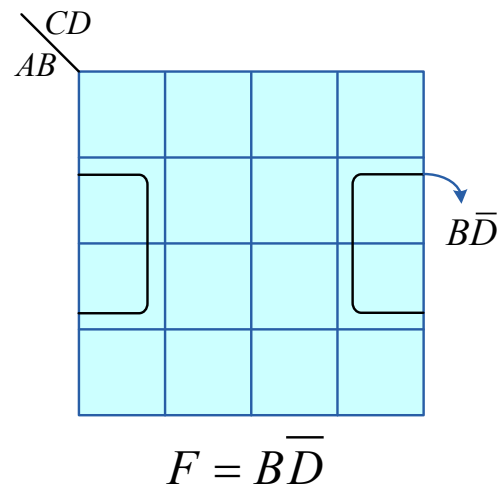
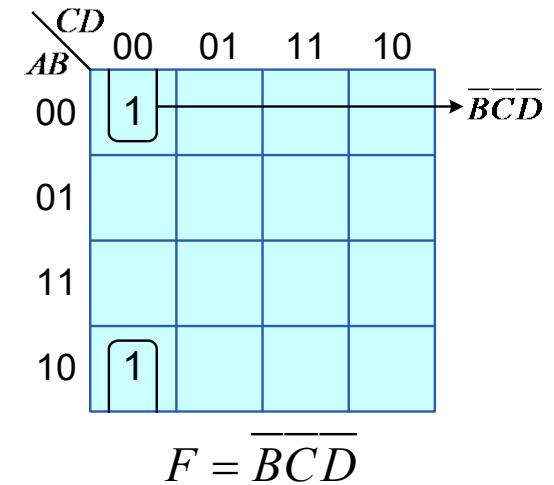
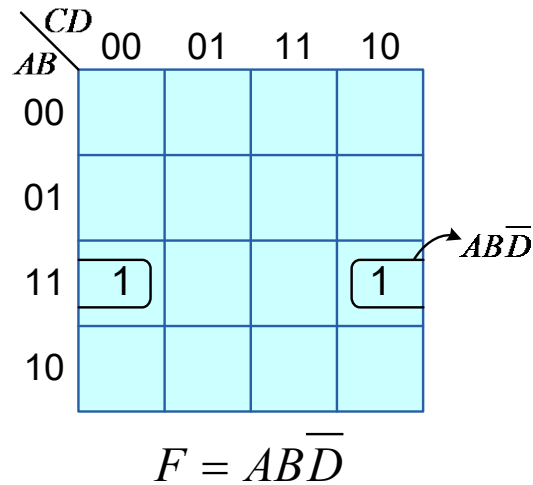
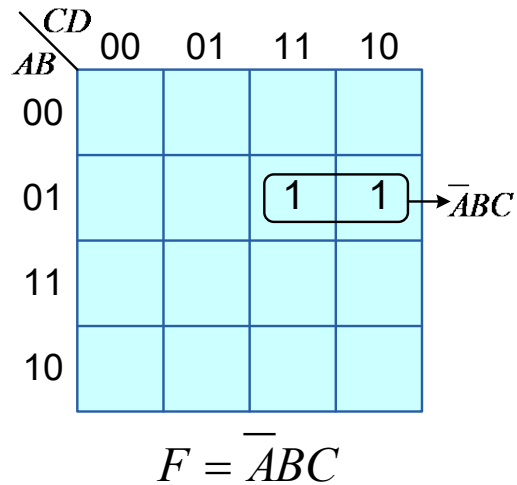
6.3 4변수 카르노 맵

4변수 카르노 맵

예제 6-3

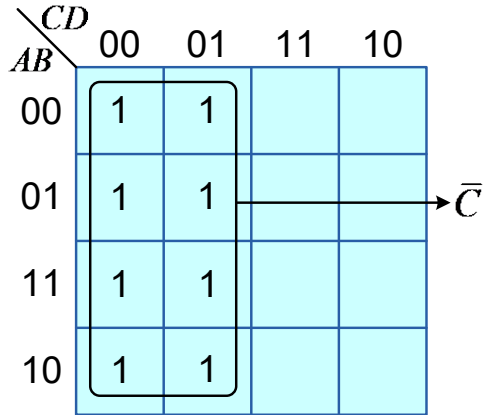
다음과 같은 다양한 4변수 카르노 맵에서 간소화된 논리식을 구하라.

풀이

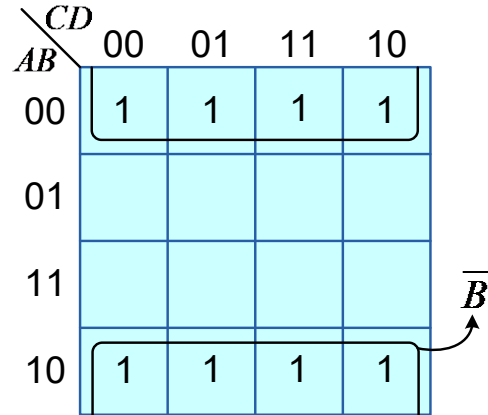


6.3 4변수 카르노 맵

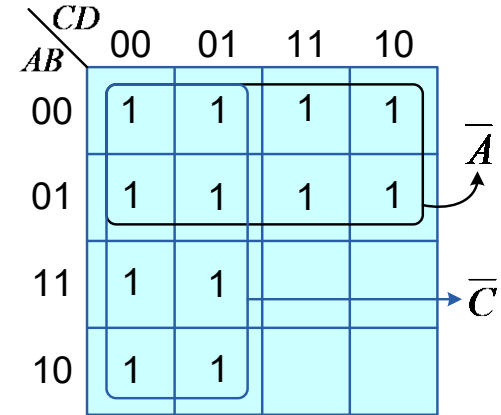
4변수 카르노 맵



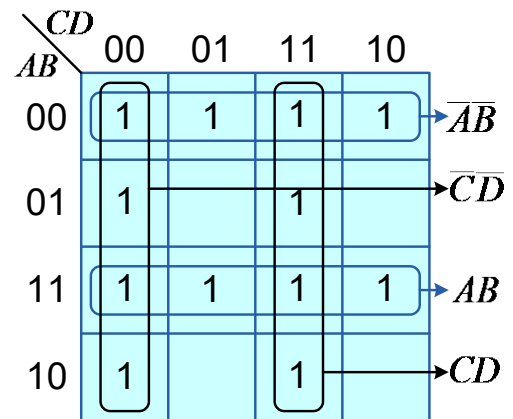
$$F = \overline{C}$$



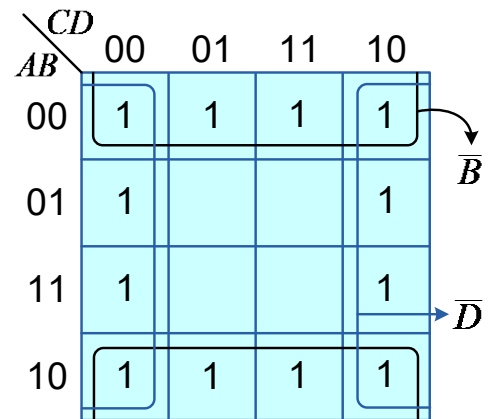
$$F = \overline{B}$$



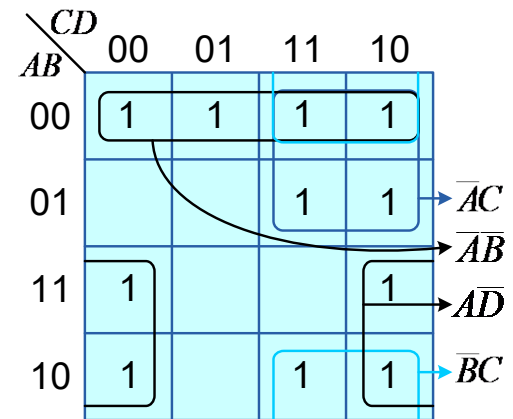
$$F = \overline{A} + \overline{C}$$



$$F = \overline{A}\overline{B} + AB + \overline{C}\overline{D} + CD$$



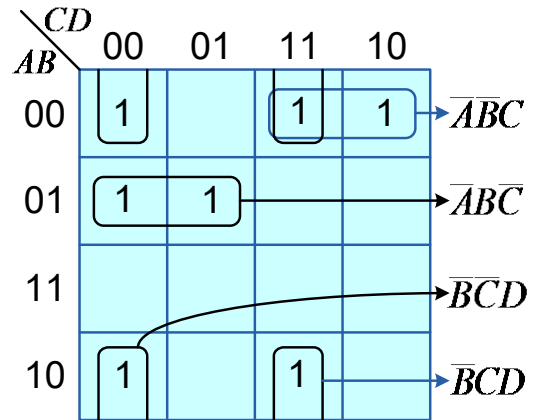
$$F = \overline{B} + \overline{D}$$



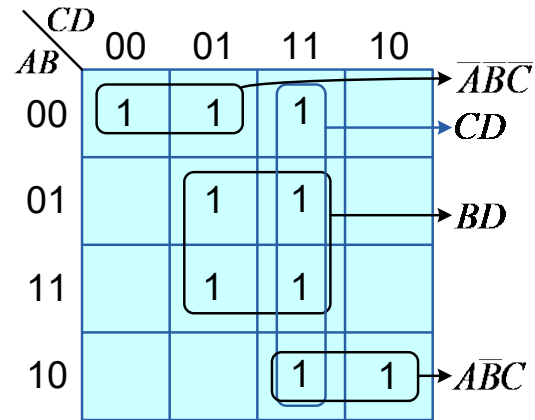
$$F = \overline{A}\overline{B} + \overline{A}\overline{D} + \overline{A}C + \overline{B}C$$

6.3 4변수 카르노 맵

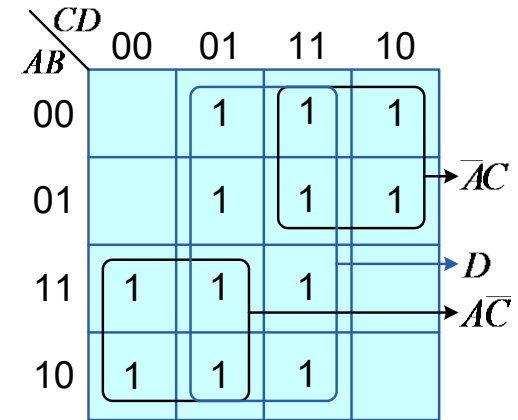
4변수 카르노 맵



$$F = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}\overline{C} + \overline{B}\overline{C}\overline{D} + \overline{B}C\overline{D}$$



$$F = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + BD + CD$$



$$F = \overline{A}\overline{C} + \overline{A}C + D$$

End of Example

예제 6-4

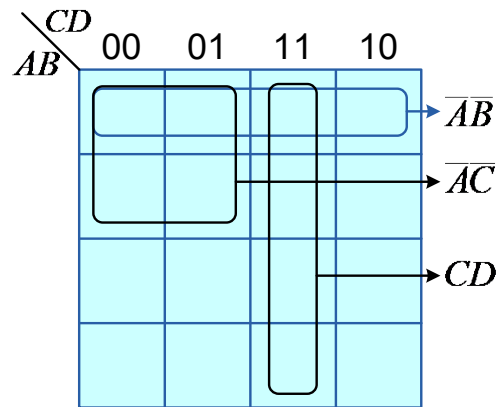
다음 식과 같이 무관항이 있을 경우, 카르노 맵을 이용해 간소화하라.

(a) $F(A,B,C,D) = \Sigma m(0,2,3,4,5,11) + \Sigma d(1,7,15)$

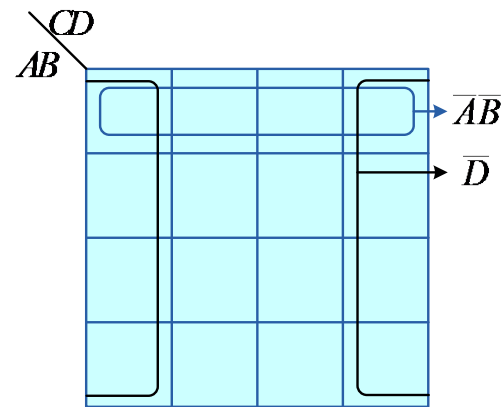
(b) $F(A,B,C,D) = \Sigma m(1,2,3,4,6,8,10) + \Sigma d(0,12,14)$

(c) $F(A,B,C,D) = \Sigma m(0,2,3,4,8,9,11) + \Sigma d(1,5,6,7,10,12)$

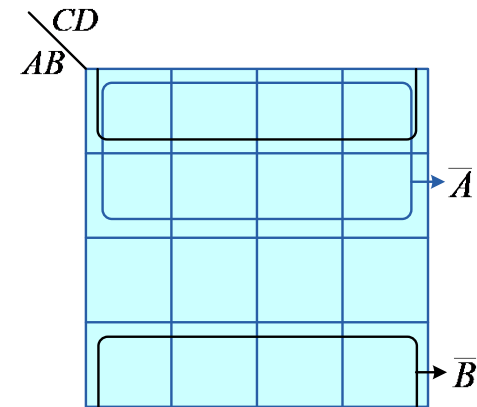
풀이



(a) $F = \overline{A}\overline{B} + \overline{A}\overline{C} + CD$



(b) $F = \overline{A}\overline{B} + \overline{D}$



(c) $F = \overline{A} + \overline{B}$

End of Example

6.3 4변수 카르노 맵

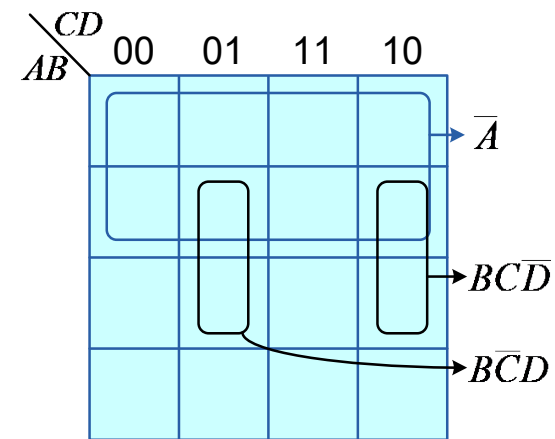
4변수 카르노 맵

예제 6-5

다음 진리표로부터 카르노 맵을 작성하고 간소화하라.

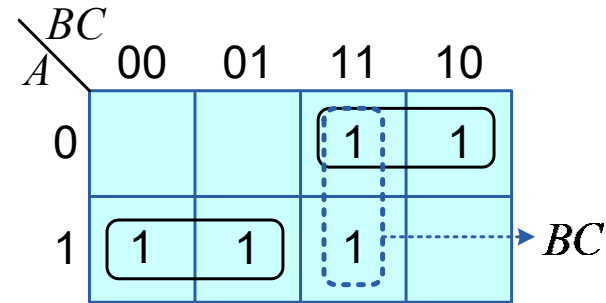
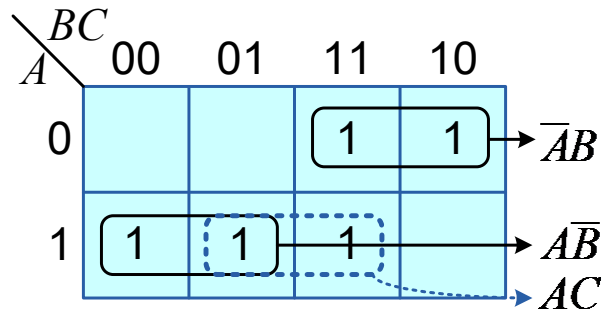
A	B	C	D	F
0	0	0	0	x
0	0	0	1	1
0	0	1	0	x
0	0	1	1	1
0	1	0	0	x
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

풀이

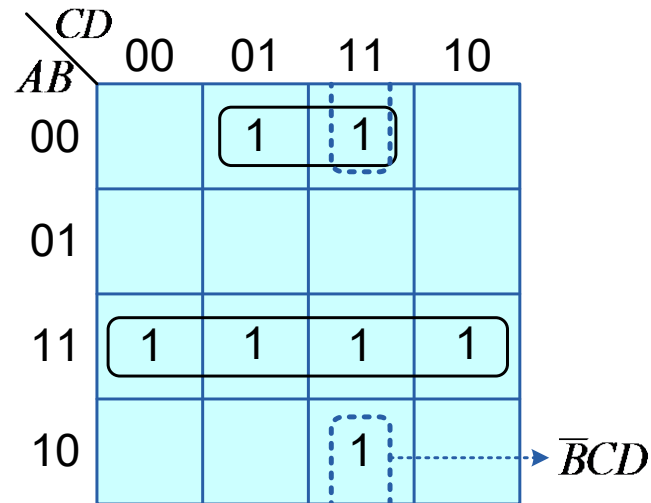
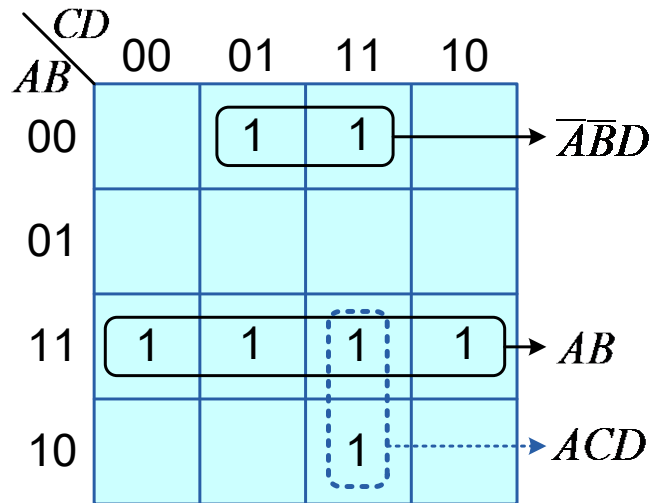


$$F(A, B, C, D) = \bar{A} + \bar{B}\bar{C}\bar{D} + \bar{B}C\bar{D}$$

□ 카르노 맵에서 선택적으로 묶을 수 있는 경우

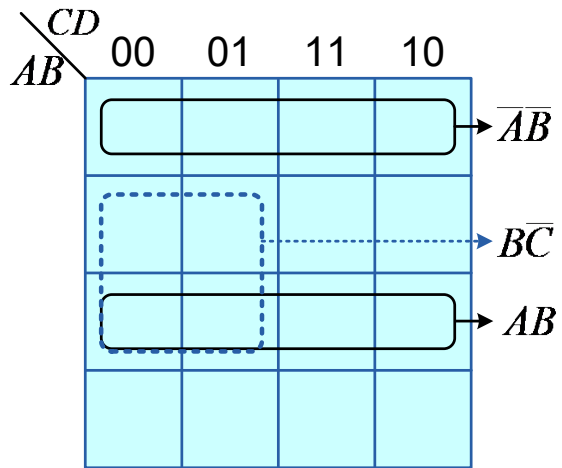


2가지 답이 가능한 경우

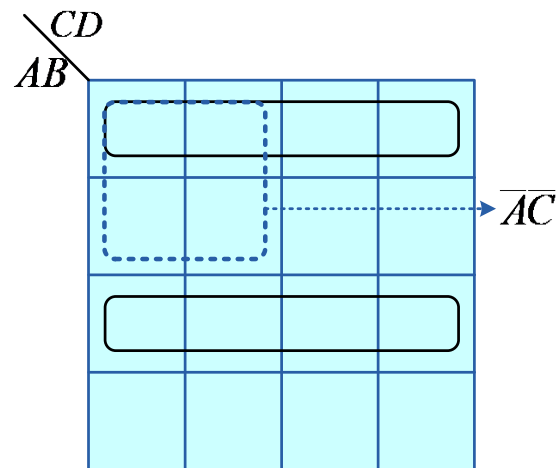


6.4 선택적 카르노 맵

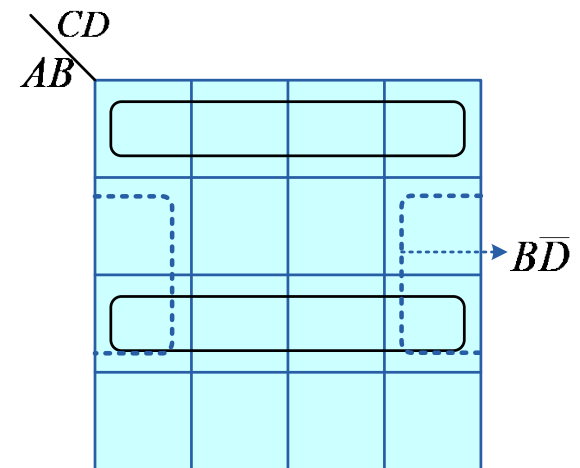
선택적 카르노 맵



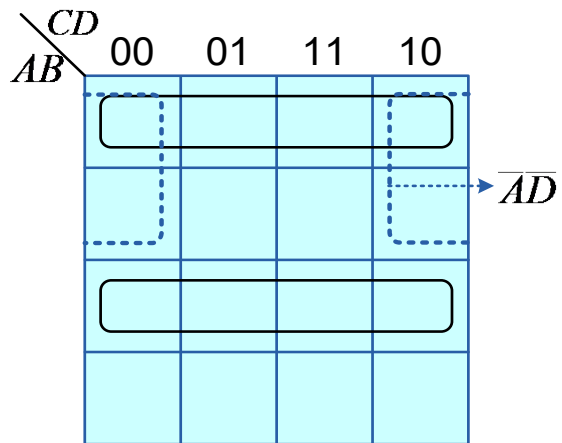
$$F = \overline{A}\overline{B} + AB + B\overline{C}$$



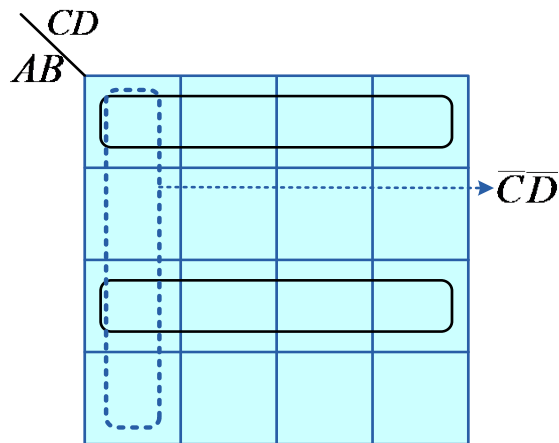
$$F = \overline{A}\overline{B} + AB + \overline{A}\overline{C}$$



$$F = \overline{A}\overline{B} + AB + B\overline{D}$$



$$F = \overline{A}\overline{B} + AB + \overline{A}\overline{D}$$



$$F = \overline{A}\overline{B} + AB + \overline{C}\overline{D}$$

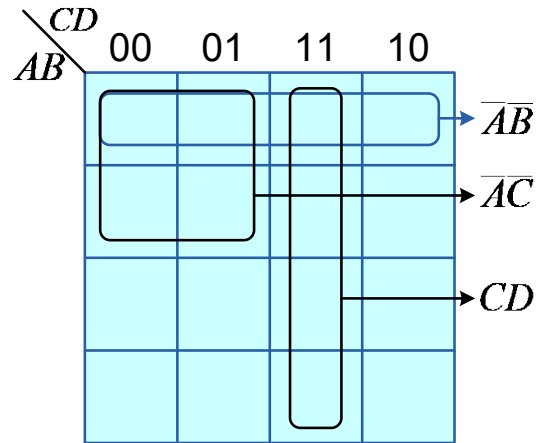
5가지 답이 가능한 경우

예제 6-6

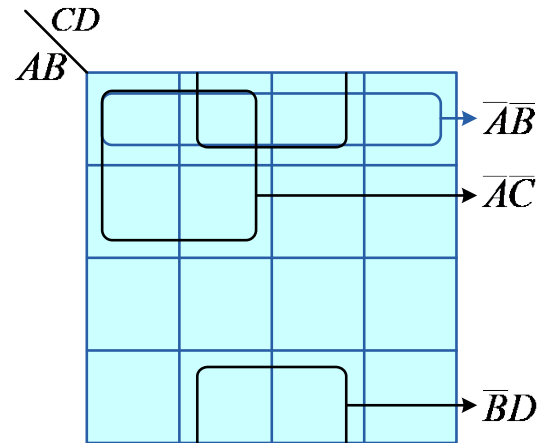
다음 논리식과 같이 무관항이 있을 경우, 2가지 선택적 카르노 맵으로 간소화하라.

$$F(A, B, C, D) = \sum m(0, 2, 3, 4, 5, 11) + \sum d(1, 7, 9, 15)$$

풀이



$$F = \overline{AB} + \overline{AC} + CD$$



$$F = \overline{AB} + \overline{AC} + \overline{BD}$$

End of Example

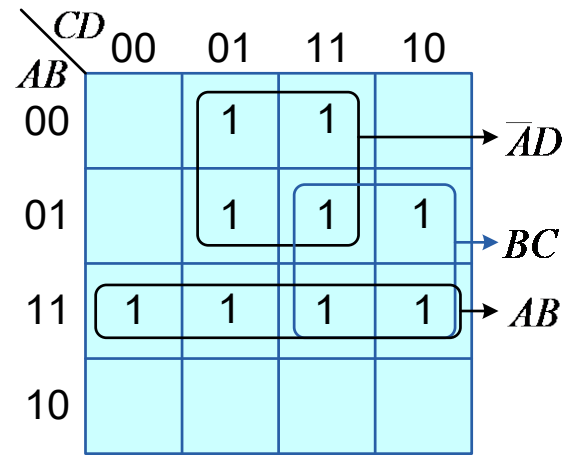
- ❖ 논리식에서 생략된 부분을 찾아서 최소항(Minterm)으로 변경
 - 변수가 없는 항을 모두 나타나도록 변경한다.

$$\begin{aligned}
 F(A, B, C) &= ABC + \overline{A}B + \overline{A}\overline{B} \\
 &= ABC + \overline{A}B(\overline{C} + C) + \overline{A}\overline{B}(\overline{C} + C) \\
 &= ABC + \overline{A}B\overline{C} + \overline{A}BC + \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C \\
 &= \overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C} + \overline{A}BC + ABC \\
 &= \Sigma m(0, 1, 2, 3, 7)
 \end{aligned}$$

$\begin{array}{c} BC \\ \diagdown A \end{array}$	00	01	11	10
0	1	1	1	1
1			1	

$$F = \overline{A} + BC$$

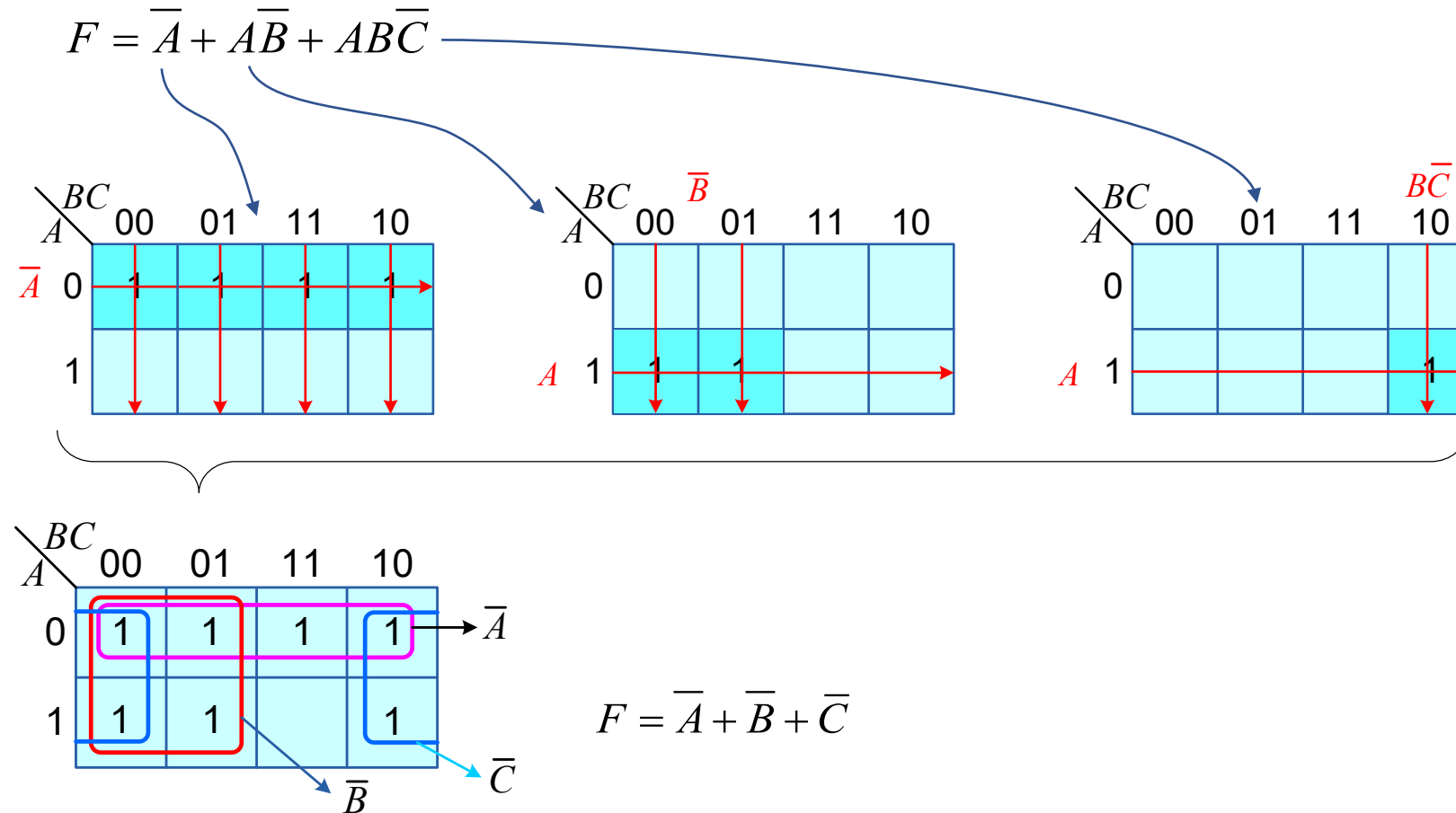
$$\begin{aligned}
 F(A, B, C, D) &= AB + ABC + \bar{A}CD + \bar{A}\bar{C}D + \bar{A}BC\bar{D} \\
 &= AB(\bar{C} + C)(\bar{D} + D) + ABC(\bar{D} + D) + \bar{A}(\bar{B} + B)CD + \bar{A}(\bar{B} + B)\bar{C}D + \bar{A}BC\bar{D} \\
 &= (AB\bar{C} + ABC)(\bar{D} + D) + ABC\bar{D} + ABCD + \bar{A}\bar{B}CD + \bar{A}BCD \\
 &\quad + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} \\
 &= AB\bar{C}\bar{D} + AB\bar{C}D + ABC\bar{D} + ABCD + ABC\bar{D} + ABCD + \bar{A}\bar{B}CD \\
 &\quad + \bar{A}\bar{B}CD + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} \\
 &= \Sigma m(12, 13, 14, 15, 3, 7, 1, 5, 6) = \Sigma m(1, 3, 5, 6, 7, 12, 13, 14, 15) \\
 &= AB + \bar{A}D + BC
 \end{aligned}$$



$$F = AB + \bar{A}D + BC$$

6.5 논리식의 카르노 맵 작성

❖ 최소항식으로 전개하지 않고 직접 카르노 맵을 이용하는 방법



6.5 논리식의 카르노 맵 작성

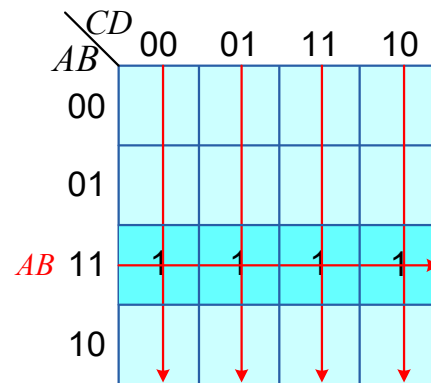
논리식의 카르노 맵 작성

예제 6-7

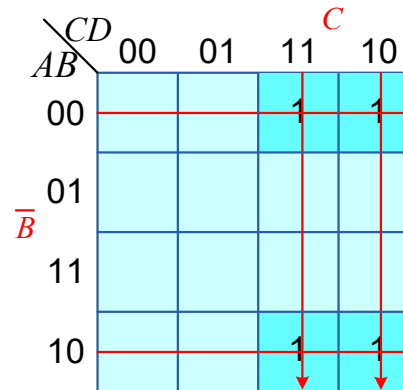
다음 논리식은 변수 4개로 구성되어 있고, 모든 항이 이미 어느 정도 간소화된 상태다. 더 간소화할 수 있는지 카르노 맵을 이용하여 확인하라.

$$F = AB + \bar{B}C + ACD + AB\bar{D} + ACD\bar{D}$$

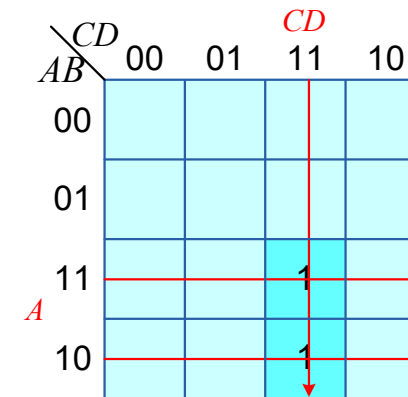
풀이



(a) AB

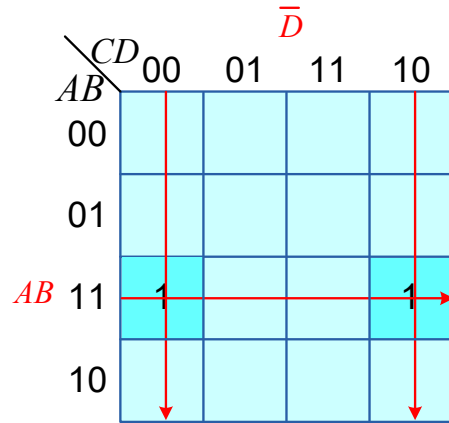


(b) $\bar{B}C$

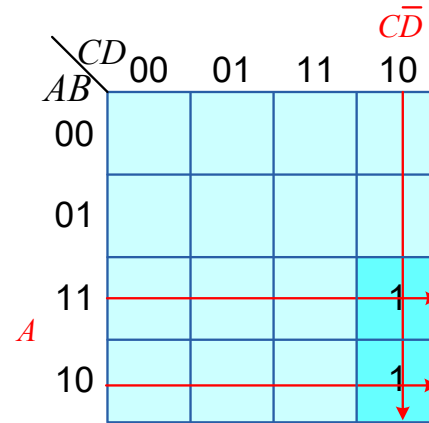


(c) ACD

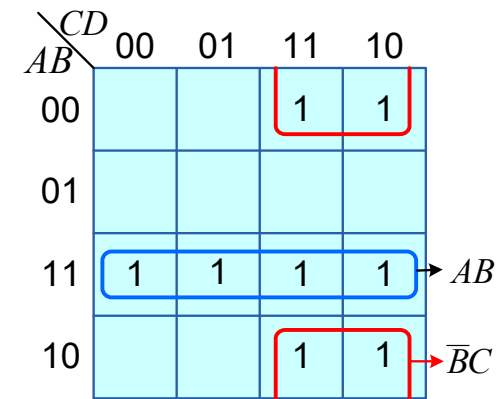
$$F = AB + \bar{B}C + ACD + AB\bar{D} + ACD\bar{D}$$



(d) $AB\bar{D}$



(e) $ACD\bar{D}$

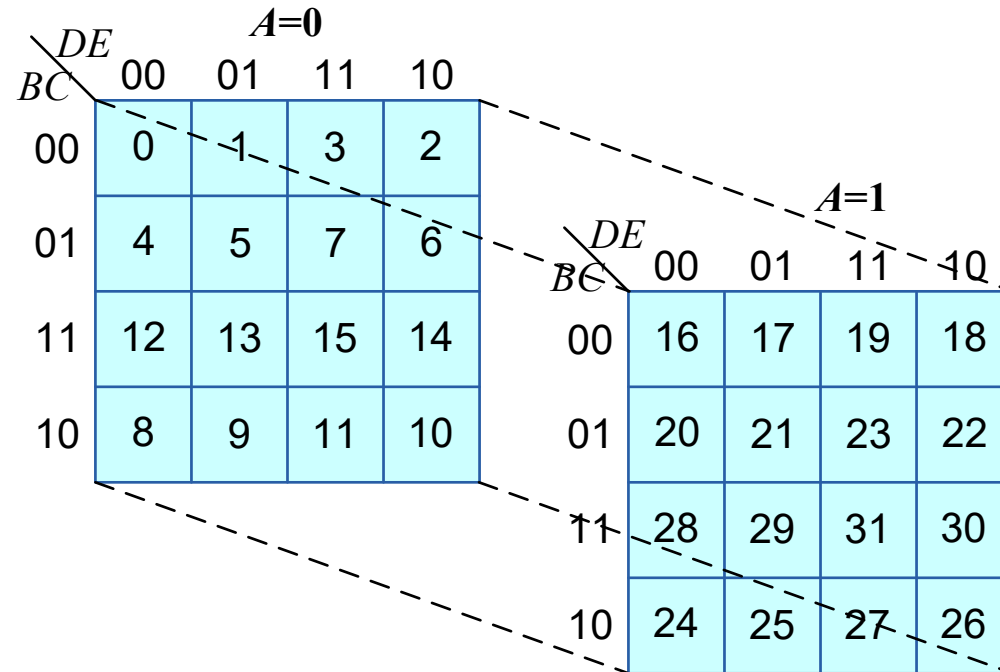


(f) $F = AB + \bar{B}C$

(a)+(b)+(c)+(d)+(e)

End of Example

□ 5변수인 경우



5변수 카르노 맵

6.6 5변수, 6변수 카르노 맵

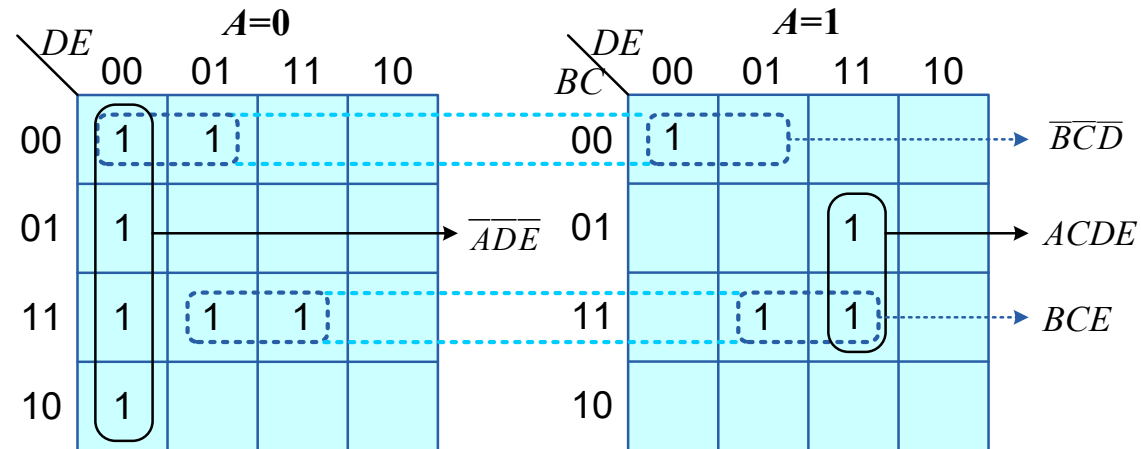
5변수 카르노 맵

예제 6-8

다음 5변수 논리함수를 카르노 맵을 이용하여 간소화하라.

$$F(A, B, C, D, E) = \sum m(4, 5, 6, 7, 9, 11, 13, 15, 16, 18, 27, 28, 31)$$

풀이



$$F = \overline{ADE} + \overline{BCD} + BCE + ACDE$$

End of Example

□ 6변수인 경우

$AB=00$					$AB=01$					$AB=11$					$AB=10$				
EF					EF					EF					EF				
CD	00	01	11	10	CD	00	01	11	10	CD	00	01	11	10	CD	00	01	11	10
00	0	1	3	2	00	16	17	19	18	00	48	49	51	50	00	32	33	35	34
01	4	5	7	6	01	20	21	23	22	01	52	53	55	54	01	36	37	39	38
11	12	13	15	14	11	28	29	31	30	11	60	61	63	62	11	44	45	47	46
10	8	9	11	10	10	24	25	27	26	10	56	57	59	58	10	40	41	43	42

6변수 카르노 맵

6.6 5변수, 6변수 카르노 맵

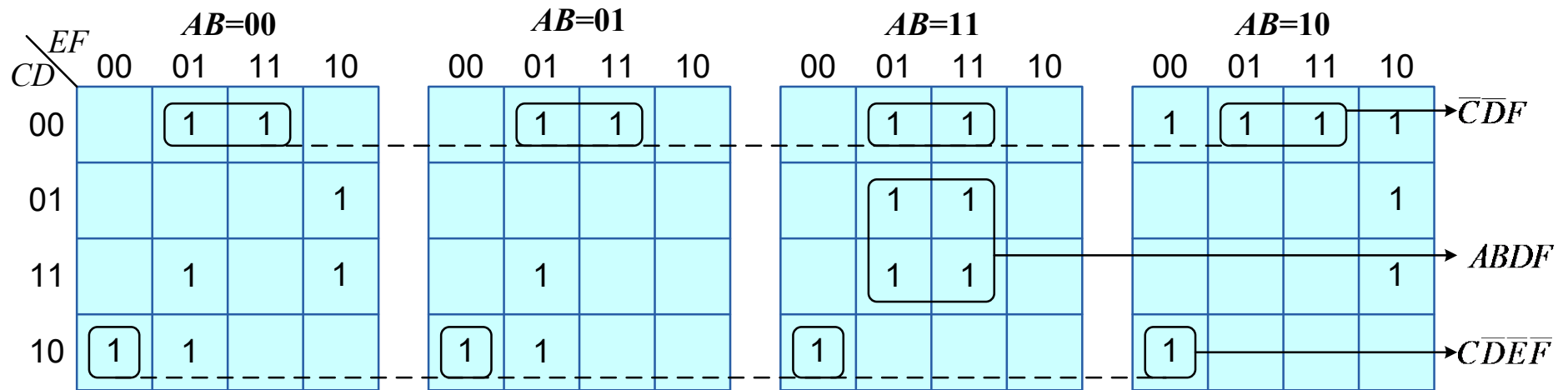
6변수 카르노 맵

예제 6-9

다음 6변수 논리함수를 카르노 맵을 이용하여 간소화하라.

$$F(A, B, C, D, E, F) = \Sigma m(1, 3, 6, 8, 9, 13, 14, 17, 19, 24, 25, 29, 32, 33, 34, 35, 38, 40, 46, 49, 51, 53, 55, 56, 61, 63)$$

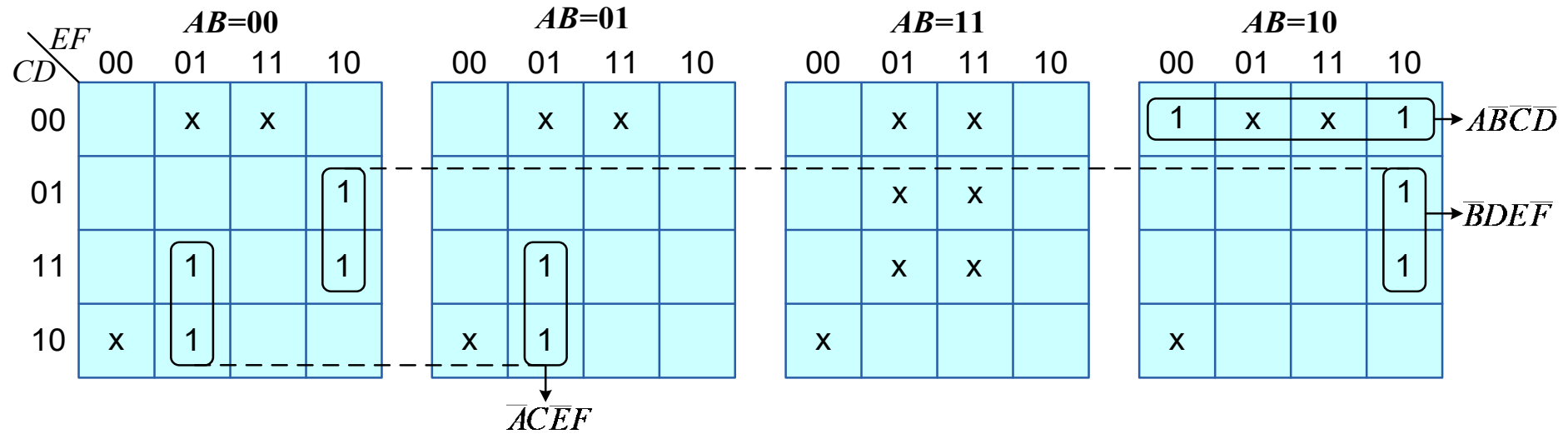
풀이



$$F_1 = ABDF + C\overline{D}\overline{E}\overline{F} + \overline{C}DF$$

6.6 5변수, 6변수 카르노 맵

6변수 카르노 맵



$$F_2 = \overline{A}CE\overline{F} + \overline{B}DE\overline{F} + \overline{A}BC\overline{D}$$

$$F = F_1 + F_2 = ABDF + C\overline{D}\overline{E}\overline{F} + \overline{C}\overline{D}F + \overline{A}CE\overline{F} + \overline{B}DE\overline{F} + \overline{A}BC\overline{D}$$

End of Example

6.7 NAND와 NOR 게이트로의 변환

NAND와 NOR 게이트로의 변환

□ 기본 게이트의 NAND, NOR 식

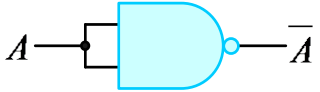
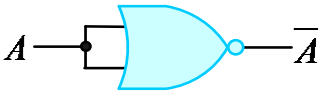
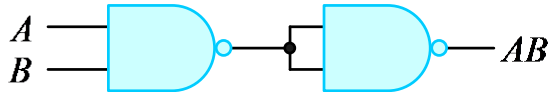
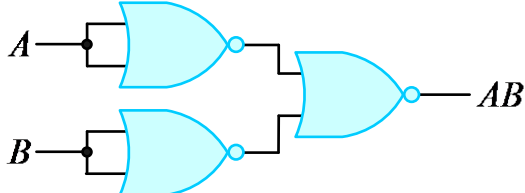
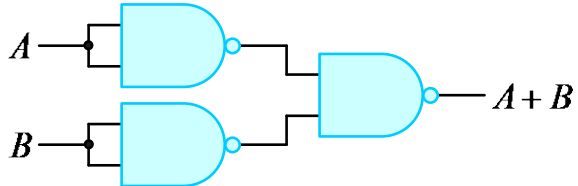
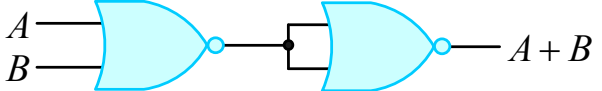
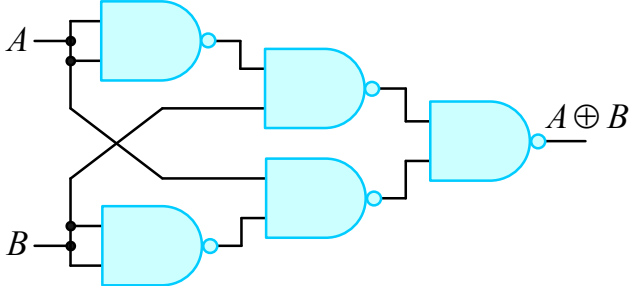
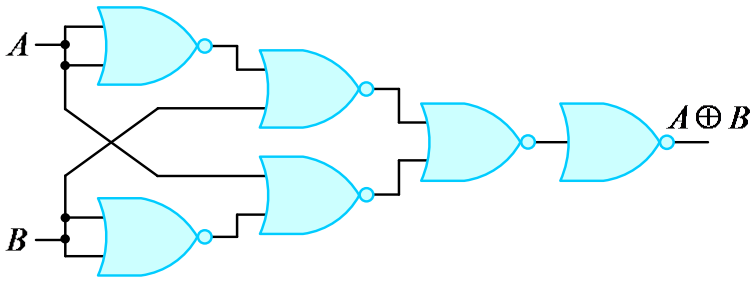
NOT	$\overline{A} = \overline{A + A} = \overline{A \cdot A}$
AND	$AB = \overline{\overline{AB}} = \overline{\overline{A} + \overline{B}}$
OR	$A + B = \overline{\overline{A + B}} = \overline{\overline{A} \cdot \overline{B}}$
NAND	$\overline{AB} = \overline{\overline{\overline{AB}}} = \overline{\overline{A} + \overline{B}}$
NOR	$\overline{A + B} = \overline{\overline{\overline{A + B}}} = \overline{\overline{A} \cdot \overline{B}}$
XOR	$\overline{AB} + \overline{AB} = \overline{\overline{\overline{AB} + \overline{AB}}} = \overline{\overline{AB} \cdot \overline{AB}} = \overline{(A + \overline{B})(\overline{A} + B)}$ $= \overline{\overline{(A + \overline{B})} + \overline{(\overline{A} + B)}}$

TIP

universal gate

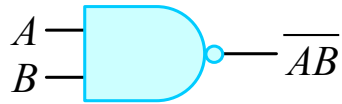
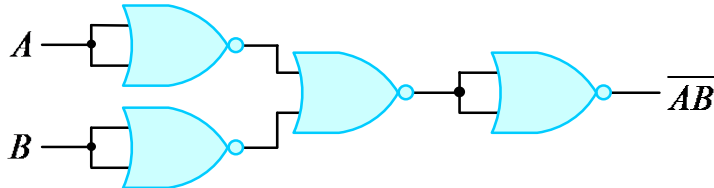
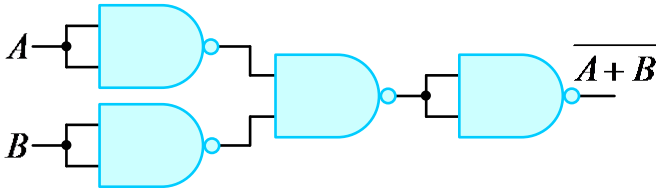
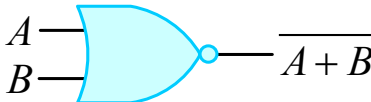
NAND와 NOR 게이트는 만능 게이트라고도 불린다. 이 두 게이트는 다른 어떤 종류의 게이트라도 구성할 수 있기 때문이다.

□ 기본 게이트의 NAND, NOR 회로

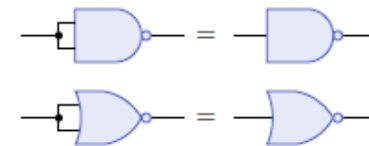
기본 게이트	NAND 게이트로 표현	NOR 게이트로 표현
NOT		
AND		
OR		
XOR		

6.7 NAND와 NOR 게이트로의 변환

NAND와 NOR 게이트로의 변환

기본 게이트	NAND 게이트로 표현	NOR 게이트로 표현
NAND		
NOR		

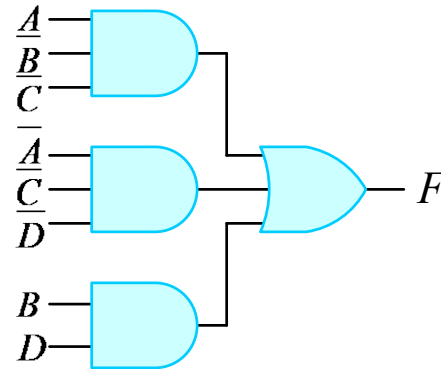
TIP



1 NAND 게이트만 이용한 회로

CD \ AB	00	01	11	10
00	1			
01	1	1	1	
11		1	1	
10	1	1		

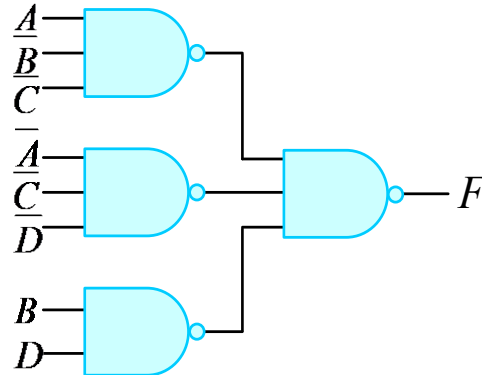
$\overline{A}\overline{C}\overline{D}$ (from AB=00, CD=00)
 BD (from AB=01, CD=01; AB=11, CD=01; AB=11, CD=11)
 ABC (from AB=10, CD=10)



$$F = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{C}\overline{D} + BD$$

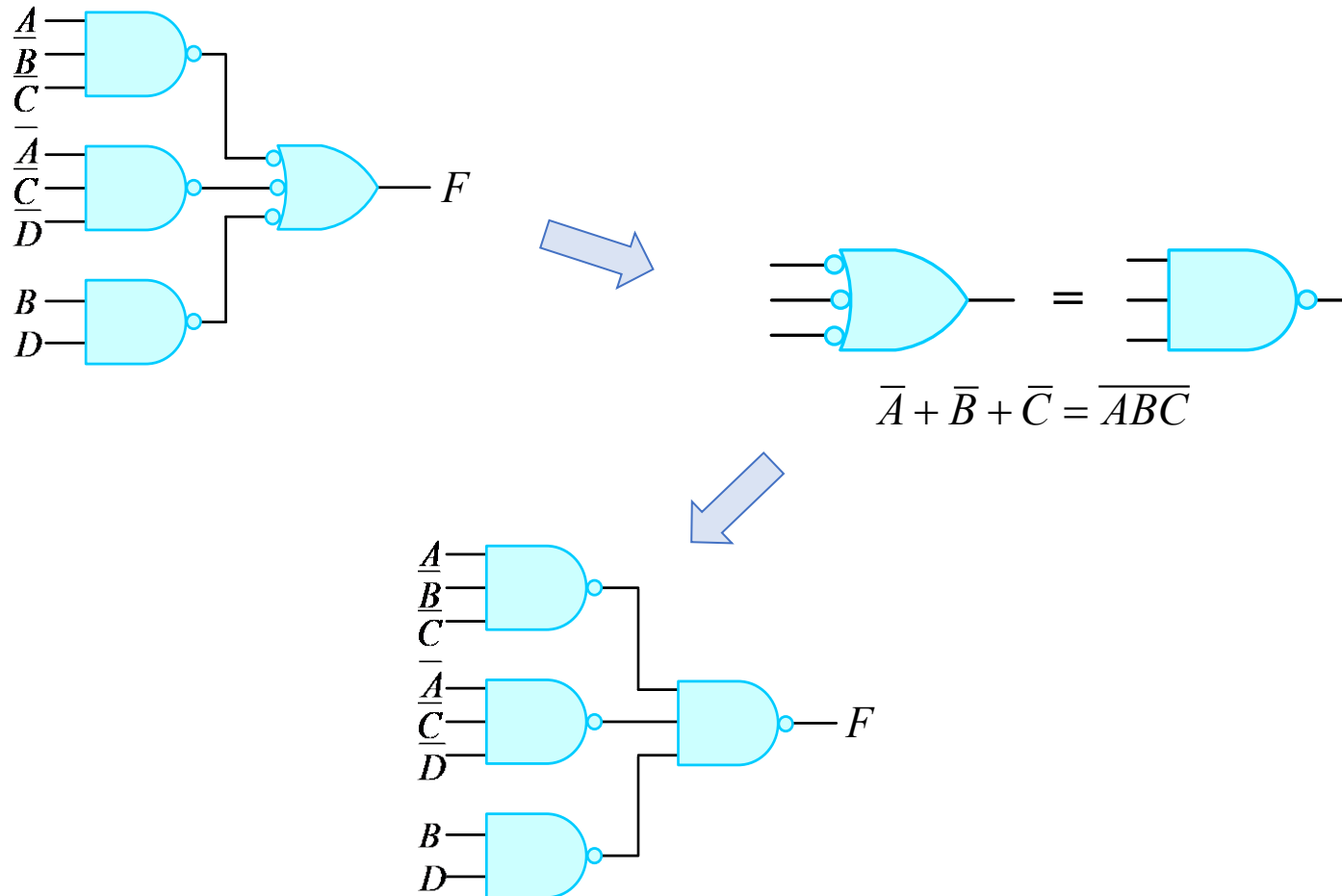
- 이 논리식을 이중부정을 하여 드모르간의 정리를 적용하여 변형

$$\begin{aligned}
 F &= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{C}\overline{D} + BD \\
 &= \overline{\overline{\overline{A}\overline{B}\overline{C}} + \overline{\overline{\overline{A}\overline{C}\overline{D}} + \overline{\overline{BD}}}} \\
 &= \overline{\overline{A}\overline{B}\overline{C} \cdot \overline{A}\overline{C}\overline{D} \cdot BD}
 \end{aligned}$$



□ 다른 방법

- 원래의 논리회로를 그린 다음, AND 게이트 뒤에 OR 게이트가 있을 때 이중부정 적용



TIP

버블 bubble

NOT 게이트를 간단하게 표현하는 동그라미 모양의 기호를 버블(기포)이라고 한다.

6.7 NAND와 NOR 게이트로의 변환

NAND 게이트만 이용한 회로

예제 6-10

다음 논리식을 NAND 게이트만 사용해 설계하라.

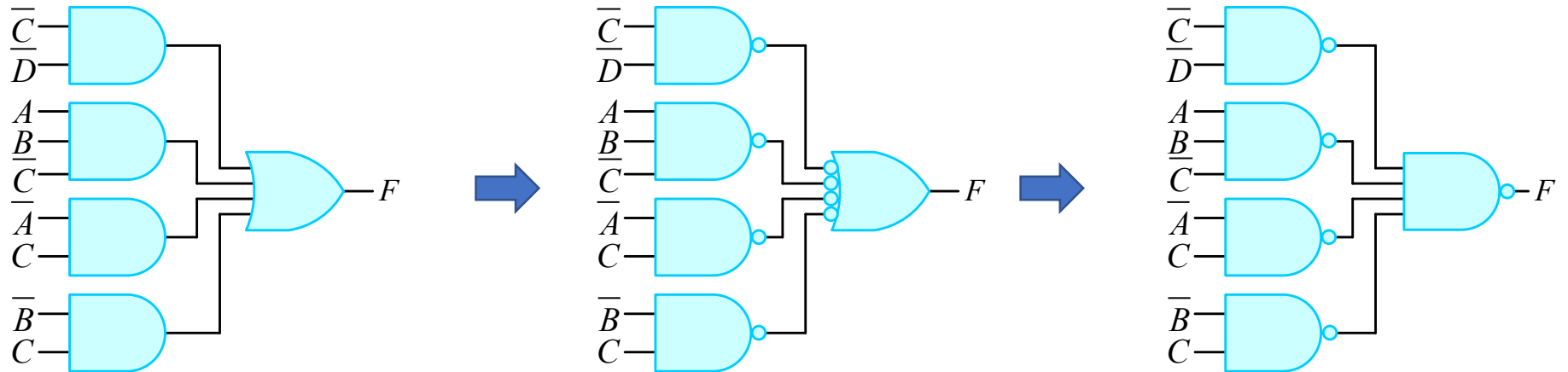
$$F = \overline{C}\overline{D} + ABC\overline{C} + \overline{A}C + \overline{B}C$$

풀이

$$F = \overline{C}\overline{D} + ABC\overline{C} + \overline{A}C + \overline{B}C$$

$$= \overline{\overline{\overline{C}\overline{D} + ABC\overline{C} + \overline{A}C + \overline{B}C}}$$

$$= \overline{\overline{C}\overline{D}} \overline{ABC\overline{C}} \overline{\overline{A}C} \overline{\overline{B}C}$$



End of Example

2 NOR 게이트만 이용한 회로

- 0을 묶어서 SOP식으로 나타내면

CD \ AB	00	01	11	10	
00	1	0	0	0	$B + \bar{C}$
01	1	1	1	0	$A + B + \bar{D}$
11	0	1	1	0	$\bar{C} + D$
10	1	1	0	0	$\bar{A} + \bar{B} + D$

0을 묶으면

$$\bar{F} = \bar{B}C + C\bar{D} + \bar{A}\bar{B}D + AB\bar{D}$$

2중 부정을 하고 풀면 POS식이 된다.

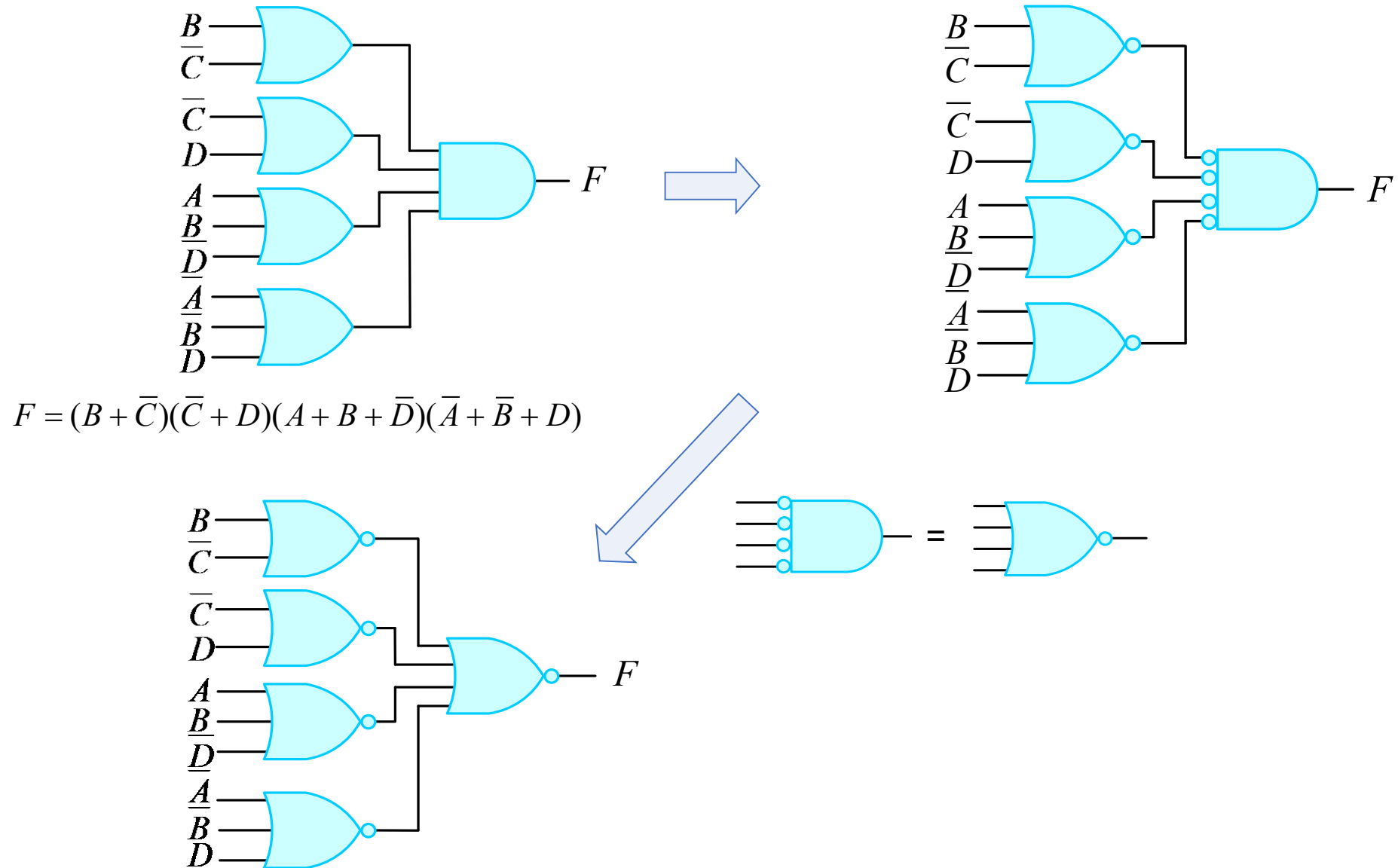
$$\begin{aligned} F &= \overline{\bar{B}C + C\bar{D} + \bar{A}\bar{B}D + AB\bar{D}} \\ &= \overline{\bar{B}C} \cdot \overline{C\bar{D}} \cdot \overline{\bar{A}\bar{B}D} \cdot \overline{AB\bar{D}} \\ &= (B + \bar{C})(\bar{C} + D)(A + B + \bar{D})(\bar{A} + \bar{B} + D) \end{aligned}$$

다시 2중 부정을 하여 나타내면 NOR회로로 직접 나타낼 수 있다.

$$\begin{aligned} \bar{\bar{F}} &= F = \overline{(B + \bar{C})(\bar{C} + D)(A + B + \bar{D})(\bar{A} + \bar{B} + D)} \\ &= \overline{(B + \bar{C}) + (\bar{C} + D) + (A + B + \bar{D}) + (\bar{A} + \bar{B} + D)} \end{aligned}$$

6.7 NAND와 NOR 게이트로의 변환

NOR 게이트만 이용한 회로



예제 6-11

다음 진리표를 만족하는 논리회로를 NOR 게이트만으로 설계하라.

<i>A B C D</i>	<i>F</i>	<i>A B C D</i>	<i>F</i>
0 0 0 0	1	1 0 0 0	0
0 0 0 1	1	1 0 0 1	0
0 0 1 0	0	1 0 1 0	1
0 0 1 1	1	1 0 1 1	1
0 1 0 0	0	1 1 0 0	0
0 1 0 1	1	1 1 0 1	1
0 1 1 0	0	1 1 1 0	0
0 1 1 1	1	1 1 1 1	1

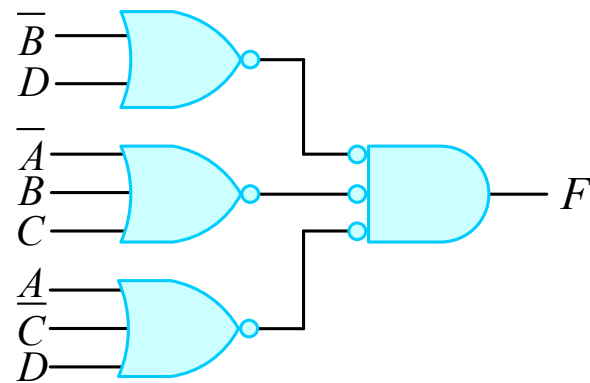
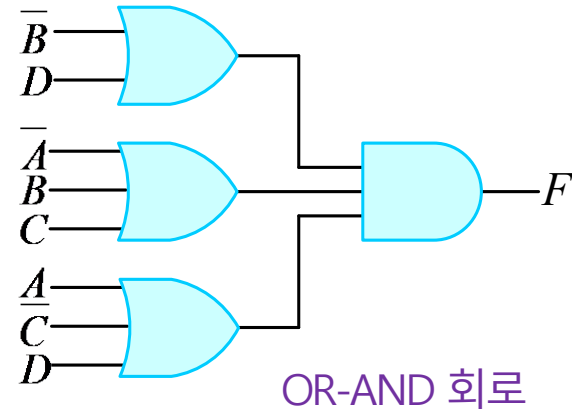
6.7 NAND와 NOR 게이트로의 변환

NOR 게이트만 이용한 회로

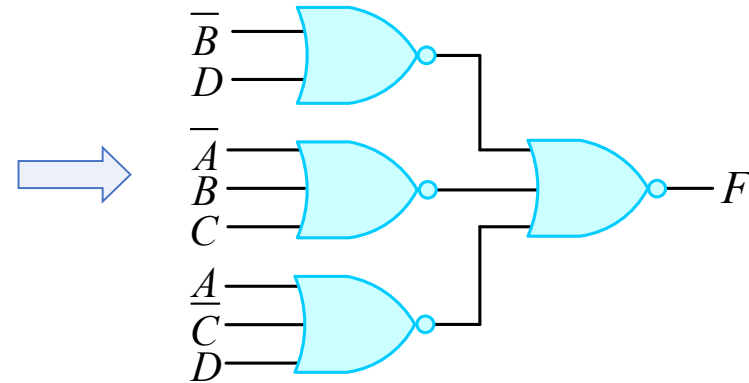
풀이 0을 묶어서 POS식으로 나타내면 $F = (\overline{B} + D)(\overline{A} + B + C)(A + \overline{C} + D)$

$\backslash CD$	00	01	11	10
AB				
00	1	1	1	0
01	0	1	1	0
11	0	1	1	0
10	0	0	1	1

$\rightarrow A + \overline{C} + D$
 $\rightarrow \overline{B} + D$
 $\rightarrow \overline{A} + B + C$



OR 출력에 이중부정



AND 부정입력을 NOR로 대체

□ XOR

- 홀수 개의 입력이 1인 경우, 출력이 1이 되는 게이트

$A \backslash BC$	00	01	11	10
0		1		1
1	1		1	

3변수 XOR

$$\begin{aligned}
 F &= \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC \\
 &= \overline{A}(\overline{B}C + B\overline{C}) + A(\overline{B}\overline{C} + BC) \\
 &= \overline{A}(B \oplus C) + A(\overline{B \oplus C}) \\
 &= A \oplus (B \oplus C) = A \oplus B \oplus C
 \end{aligned}$$

$AB \backslash CD$	00	01	11	10
00		1		1
01	1		1	
11		1		1
10	1		1	

4변수 XOR

$$\begin{aligned}
 F &= \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}BCD \\
 &\quad + A\overline{B}\overline{C}\overline{D} + A\overline{B}C\overline{D} + AB\overline{C}\overline{D} + ABC\overline{D} \\
 &= \overline{A}\overline{B}(\overline{C}D + C\overline{D}) + \overline{A}B(\overline{C}\overline{D} + CD) \\
 &\quad + A\overline{B}(\overline{C}\overline{D} + CD) + AB(\overline{C}D + C\overline{D}) \\
 &= \overline{A}\overline{B}(C \oplus D) + \overline{A}B(\overline{C \oplus D}) \\
 &\quad + A\overline{B}(\overline{C \oplus D}) + AB(C \oplus D) \\
 &= (\overline{A}\overline{B} + AB)(C \oplus D) + (\overline{A}B + A\overline{B})(\overline{C \oplus D}) \\
 &= (\overline{A \oplus B})(C \oplus D) + (A \oplus B)(\overline{C \oplus D}) \\
 &= A \oplus B \oplus C \oplus D
 \end{aligned}$$

TIP

XOR

홀함수(기함수)

XNOR

짝함수(우함수)

□ XNOR

- 짝수 개의 입력이 1인 경우 출력이 1이 되는 게이트

$\begin{array}{c} BC \\ \diagdown \end{array}$	00	01	11	10
0	1		1	
1		1		1

3변수 XNOR

$$\begin{aligned}
 F &= \overline{A}\overline{B}\overline{C} + \overline{A}BC + A\overline{B}\overline{C} + ABC\overline{C} \\
 &= \overline{A}(\overline{B}\overline{C} + BC) + A(\overline{B}\overline{C} + B\overline{C}) \\
 &= \overline{A}(\overline{B \oplus C}) + A(B \oplus C) \\
 &= \overline{A \oplus B \oplus C} = A \oplus B \oplus C
 \end{aligned}$$

$\begin{array}{c} CD \\ \diagdown \end{array}$	00	01	11	10
00	1		1	
01		1		1
11	1		1	
10		1		1

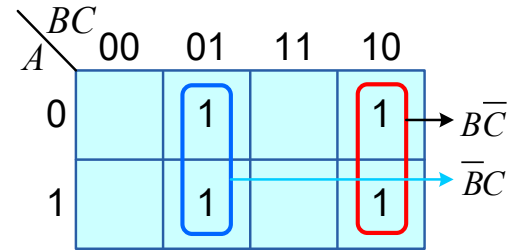
4변수 XNOR

$$\begin{aligned}
 F &= \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}CD + \overline{A}B\overline{C}\overline{D} + \overline{A}BC\overline{D} \\
 &\quad + A\overline{B}\overline{C}\overline{D} + A\overline{B}C\overline{D} + AB\overline{C}\overline{D} + ABC\overline{D} \\
 &= \overline{A}\overline{B}(\overline{C}\overline{D} + CD) + \overline{A}B(\overline{C}\overline{D} + C\overline{D}) \\
 &\quad + A\overline{B}(\overline{C}\overline{D} + C\overline{D}) + AB(\overline{C}\overline{D} + CD) \\
 &= \overline{A \oplus B \oplus C \oplus D} \\
 &= A \oplus B \oplus C \oplus D
 \end{aligned}$$

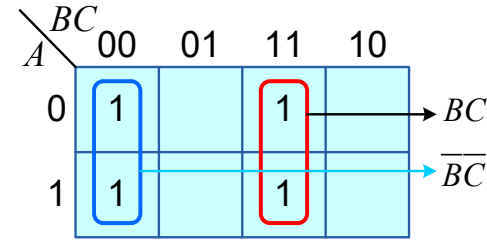
6.8 XOR와 XNOR 게이트

XOR와 XNOR 게이트

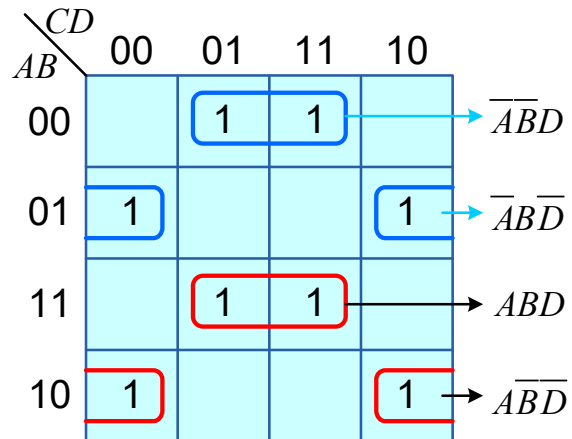
- 2개 항을 묶었을 경우 XOR 또는 XNOR 게이트의 카르노 맵 표현



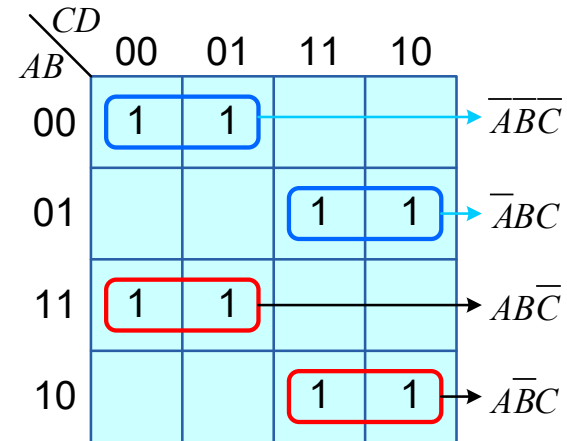
$$F = \overline{B}C + B\overline{C} = B \oplus C$$



$$F = \overline{B}\overline{C} + BC = \overline{B \oplus C} = B \square C$$



$$\begin{aligned} F &= \overline{A}\overline{B}\overline{D} + \overline{A}\overline{B}\overline{D} + \overline{A}\overline{B}\overline{D} + ABD \\ &= \overline{A}(\overline{B}\overline{D} + \overline{B}\overline{D}) + A(\overline{B}\overline{D} + BD) \\ &= \overline{A}(\overline{B} \oplus D) + A(\overline{B} \oplus D) \\ &= A \oplus B \oplus D \end{aligned}$$



$$\begin{aligned} F &= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}\overline{C} + ABC \\ &= \overline{A}(\overline{B}\overline{C} + \overline{B}\overline{C}) + A(\overline{B}\overline{C} + BC) \\ &= \overline{A}(\overline{B} \oplus C) + A(\overline{B} \oplus C) \\ &= A \oplus B \oplus C = A \square B \square C \end{aligned}$$

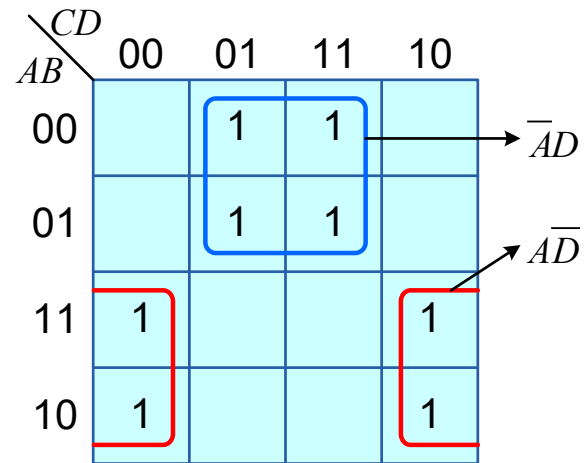
예제 6-12

다음 논리함수 F 와 G 를 XOR 게이트 또는 XNOR 게이트로 표현할 수 있는지 알아보라.

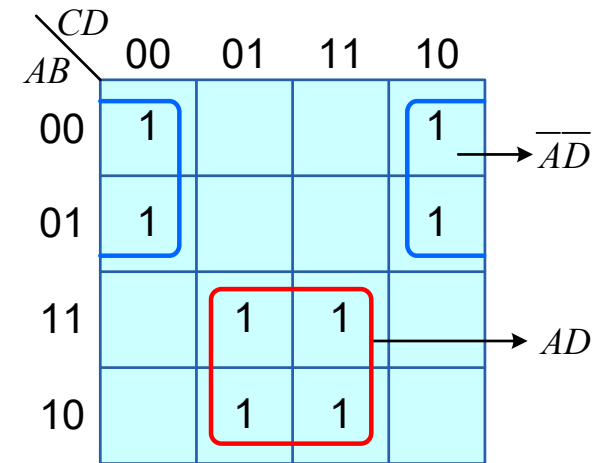
$$F(A, B, C, D) = \Sigma m(1, 3, 5, 7, 8, 10, 12, 14)$$

$$G(A, B, C, D) = \Sigma m(0, 2, 4, 6, 9, 11, 13, 15)$$

풀이



$$F = \overline{A}D + A\overline{D} = A \oplus D$$



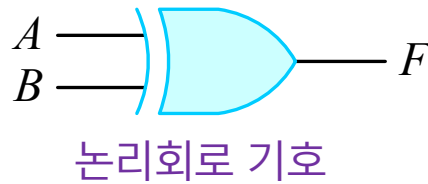
$$G = \overline{A}\overline{D} + AD = \overline{A \oplus D} = A \boxtimes D$$

End of Example

□ XOR 게이트의 다른 표현

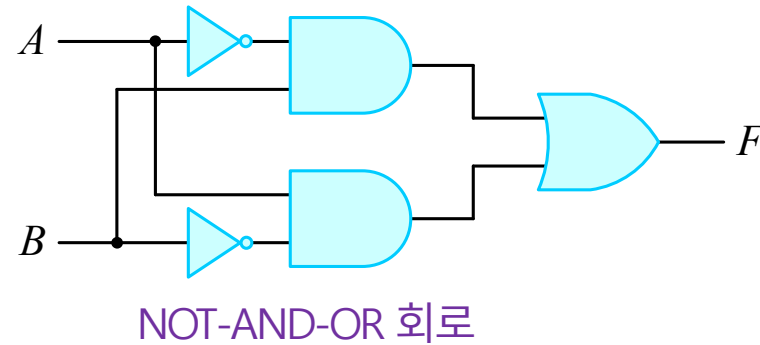
- XOR : 입력된 신호가 서로 다를 때 출력이 1이 되는 게이트

$$F = \overline{A}B + A\overline{B} = A \oplus B$$



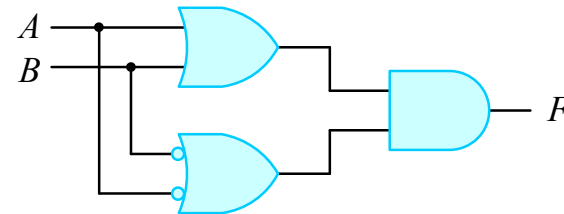
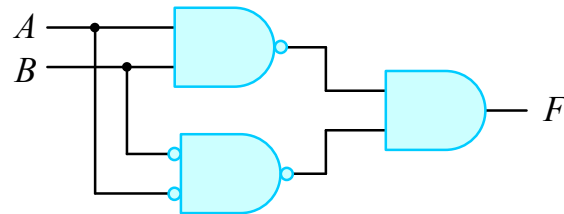
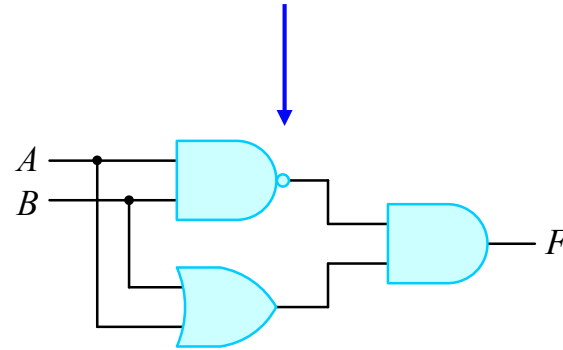
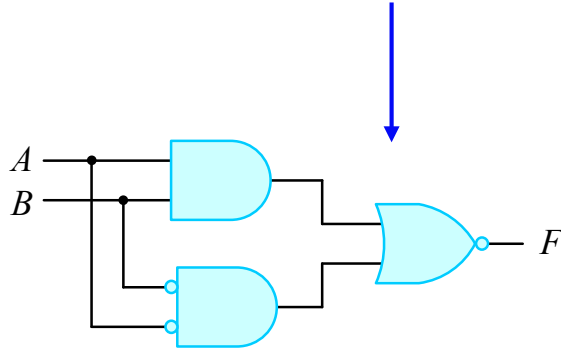
A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

진리표



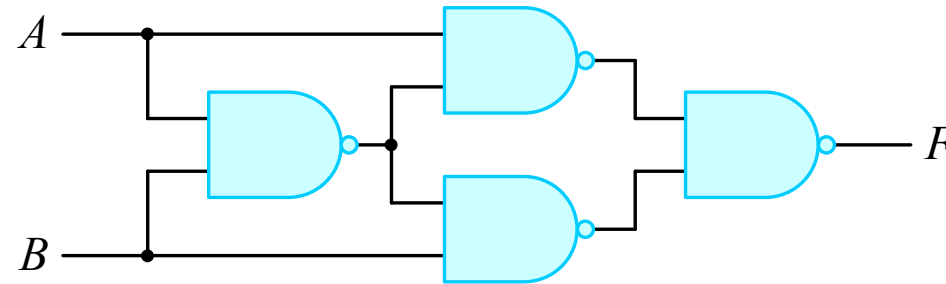
- XOR : 두 입력이 모두 0이거나 1이면 출력이 0이 되는 게이트

$$F = \overline{AB} + \overline{\overline{A}\overline{B}} = (\overline{AB})(\overline{\overline{A}\overline{B}}) = (A + B)\overline{AB} = (A + B)(\overline{A} + \overline{B})$$



- XOR를 NAND 만으로 표현하기 위하여 이중부정을 취하고 드모르간의 정리를 적용하여 정리

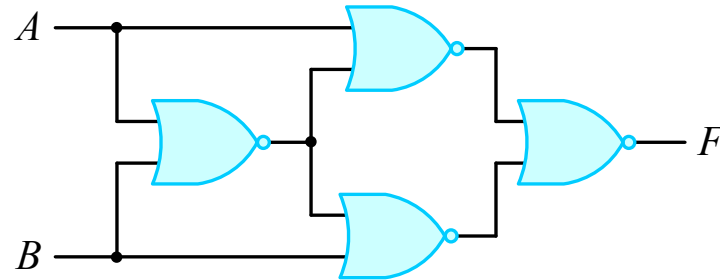
$$\begin{aligned}
 F &= \overline{AB + \overline{A}\overline{B}} = (A + B)\overline{AB} = A \cdot \overline{AB} + B \cdot \overline{AB} \\
 &= \overline{\overline{A \cdot \overline{AB} + B \cdot \overline{AB}}} = \overline{\overline{A \cdot \overline{AB}} \cdot \overline{B \cdot \overline{AB}}}
 \end{aligned}$$



XOR 게이트를 NAND 게이트만으로 표현

예제 6-13

다음의 NOR 게이트로 이루어진 회로는 어떤 게이트와 같은지 살펴보자.



풀이 논리회로로부터 논리식을 작성하고 정리하면 XNOR 게이트가 된다.

$$\begin{aligned}
 F &= \overline{\overline{A + (A + B)} + \overline{B + (A + B)}} = \overline{\overline{A + (A + B)} \cdot \overline{B + (A + B)}} \\
 &= \overline{(A + \overline{A + B}) \cdot (B + \overline{A + B})} = \overline{(A + \overline{A}\overline{B}) \cdot (B + \overline{A}\overline{B})} \\
 &= \overline{AB + \overline{A}\overline{B}} = A \oplus B
 \end{aligned}$$

End of Example